SLECS Explorer User Manual

1. Introduction

SLECS Explorer is an open-source software package for constructing and exploring Skeleton-Level Chemical Space (SLECS) from tandem mass spectrometry (MS/MS) data. The software implements the complete SLECS workflow, including optional spectrum preprocessing, fragment-matrix construction, matrix filtering, non-negative matrix factorization (NMF), SLECS generation, and interactive visualization.

The source code, demonstration datasets, and documentation are publicly available on GitHub, while all datasets and source data associated with this study are available through Zenodo.

SLECS Explorer

Skeleton-Level Chemical Space

A framework for exploring skeleton-level chemical space to enable structure-guided discovery from MS^2 data

2. Running Modes

SLECS Explorer can be used in two ways.

2.1 Online Demo (GitHub Pages)

No installation required.

Suitable for previewing the interface and visualization.

Available at: https://daffodils520.github.io/SLECS-Explorer/

The online demo allows users to browse the interface and explore example visualizations. Computational analyses and server-dependent interactive functions are available through the local deployment.

2.2 Local Deployment (Recommended)

Provides the complete functionality of SLECS Explorer.

Clone (or download) the repository and run scripts/app.py.

Recommended for preprocessing, matrix construction, NMF analysis, and fully interactive visualization.

A dedicated online server is planned in future releases to provide the complete interactive functionality through a web browser.

3. Installation

The complete source code is available from the GitHub repository.

Users may either clone the repository using Git or download the complete ZIP package directly from GitHub.

3.1 Obtain the Source Code

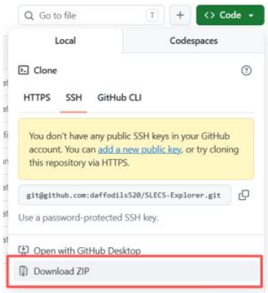
Option 1. Clone the repository

```bash
git clone https://github.com/daffodils520/SLECS-Explorer.git
cd SLECS-Explorer
```

Option 2. Download ZIP

Click Code → Download ZIP on the GitHub repository page.

After downloading, extract the ZIP package to a local directory.



3.2 First-Time Installation

Open **Anaconda Prompt** and create a dedicated environment:

```bash
conda create -n slecs_explorer python=3.11 -y
conda activate slecs_explorer
```

Enter the SLECS Explorer project directory:

```bash
cd /d E:\path\to\SLECS-Explorer
```

Replace the example path with the actual location of the downloaded project folder.

Install the required Python packages:

```bash
pip install -r requirements.txt
```

Launch SLECS Explorer:

```bash
python scripts/app.py
```

After the server starts successfully, open the local address shown in the terminal (typically http://127.0.0.1:5000) using a web browser.

```
http://127.0.0.1:5000
```

Keep the Anaconda Prompt window open while using SLECS Explorer.

3.3 Subsequent Use

The environment and dependencies only need to be installed once. For subsequent use, open **Anaconda Prompt** and run:

```bash
conda activate slecs_explorer
cd /d E:\path\to\SLECS-Explorer
python scripts/app.py
```

Then open:

```
http://127.0.0.1:5000
```

4. Input Files

SLECS Explorer accepts the following input files:

File	Description
`*.mgf`	MS/MS spectra exported from MZmine or other software
`*.graphml`	Molecular networking results generated by GNPS-FBMN (optional)
`*.csv`	Preprocessed feature table

Representative example datasets are provided in the **demo**/ and **input**/ folders.

5. Workflow Overview

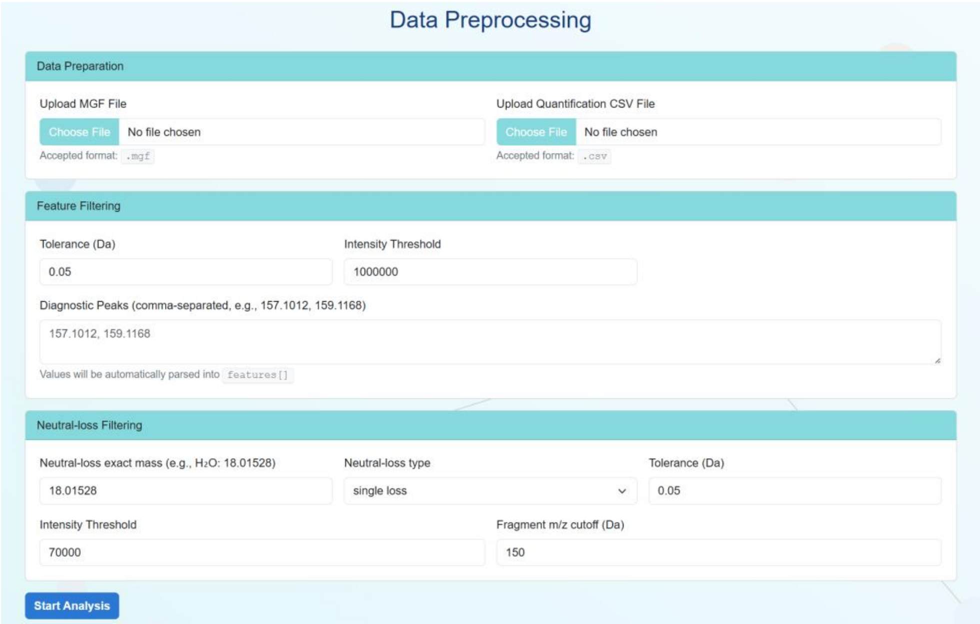
The complete workflow consists of the following steps:

① Data import
②Optional spectrum preprocessing
③Matrix construction and filtering
④NMF decomposition
⑤SLECS generation
⑥Interactive exploration and Export

6. Step-by-Step Tutorial

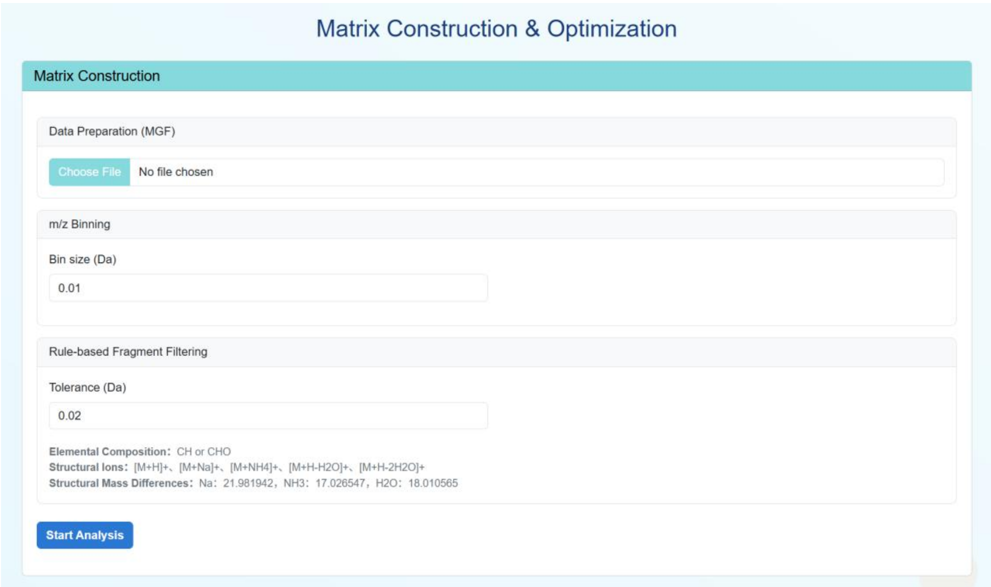
Step 1. Data Preprocessing (Optional)

Upload the MGF and CSV files, set the preprocessing parameters if required, and click Start Analysis.



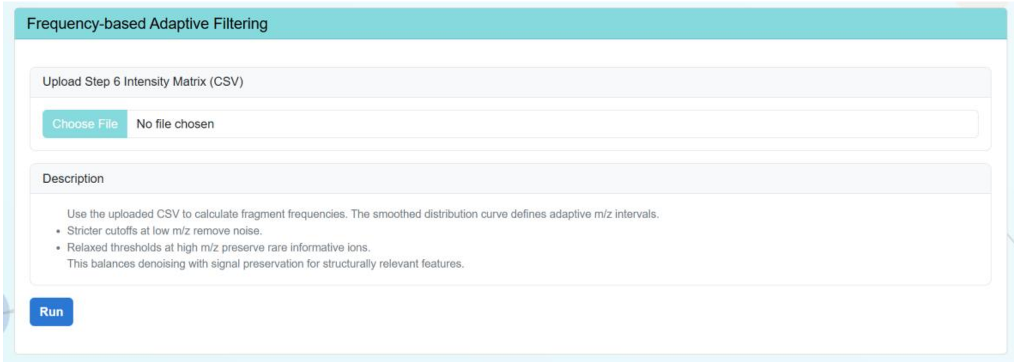
Step 2. Matrix Construction

Upload the filtered MGF file, set the matrix construction parameters, and click Start Analysis to generate the fragment-intensity matrix.



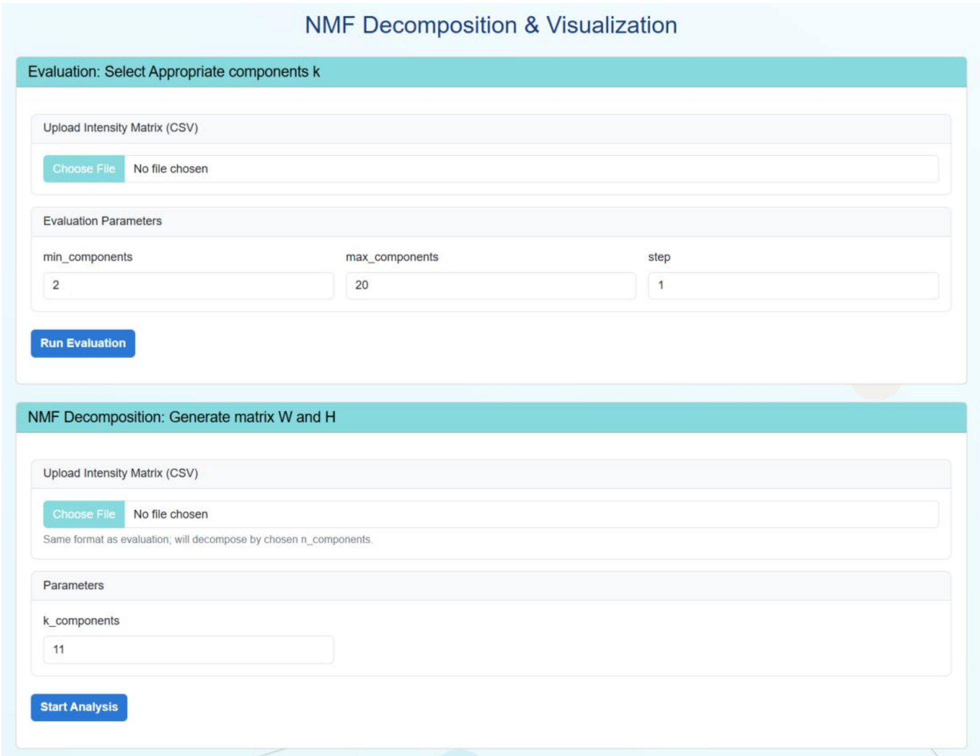
Step 3. Adaptive Filtering

Upload the intensity matrix, adjust the filtering parameters if necessary, and click Start Analysis to obtain the filtered matrix V (final matrix).



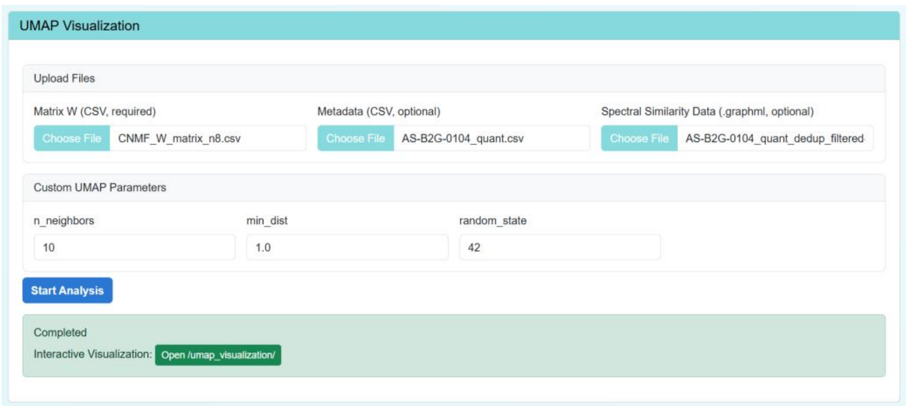
Step 4. NMF Decomposition

Upload the optimized intensity matrix, evaluate the appropriate number of components (k), and perform NMF decomposition to generate the W and H matrices.



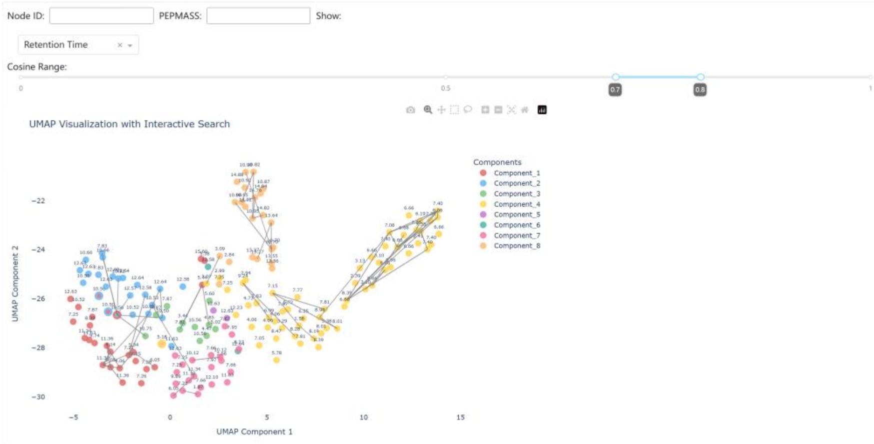
Step 5. SLECS Visualization

Upload the W matrix (metadata and molecular networking files are optional), adjust the UMAP parameters if needed, and click Start Analysis to visualize the SLECS.



Step 6. Interactive Exploration and Export

Explore the generated SLECS using the interactive interface. Users can search nodes by Node ID or PEPMASS, select the displayed node information, adjust the cosine similarity threshold interactively, overlay molecular networking, and export figures and analysis results.



7. Notes

Representative example datasets for reproducing the complete SLECS workflow are provided in the input/ directory, while the demo/ directory contains example intensity matrices for the NMF decomposition and visualization modules.

The latest source code, software updates, and documentation are available from the GitHub repository: https://github.com/daffodils520/SLECS-Explorer

The complete datasets supporting this study are publicly available through the Zenodo repository: https://doi.org/10.5281/zenodo.20700190

8. Contact

For questions regarding the SLECS Explorer software, workflow, datasets, or technical support, please contact:

Chenlan Shu: shuchenlan@simm.ac.cn

Zhuohao Yu: yuzhuohao@simm.ac.cn

9. Citation

If SLECS Explorer is used in published work, please cite the corresponding publication.